

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning,

Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

I **Savitha R / 192421368** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

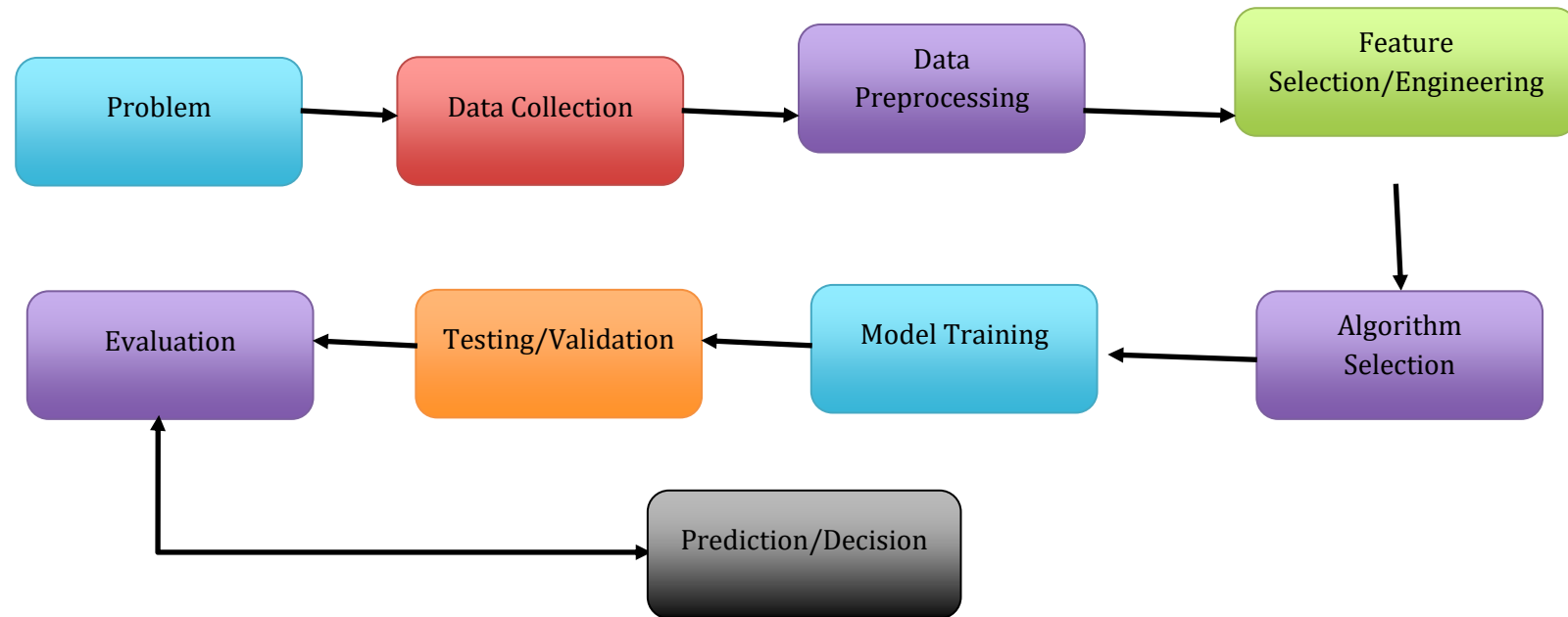
Signature of the student: SAVITHA R

Question 1

Central Concept: Machine Learning

With the necessary sketch, explain all the possible blocks in the machine-learning system. Give an example showing the concept of supervised and unsupervised learning methods in the machine learning scheme.

Flow Chart :

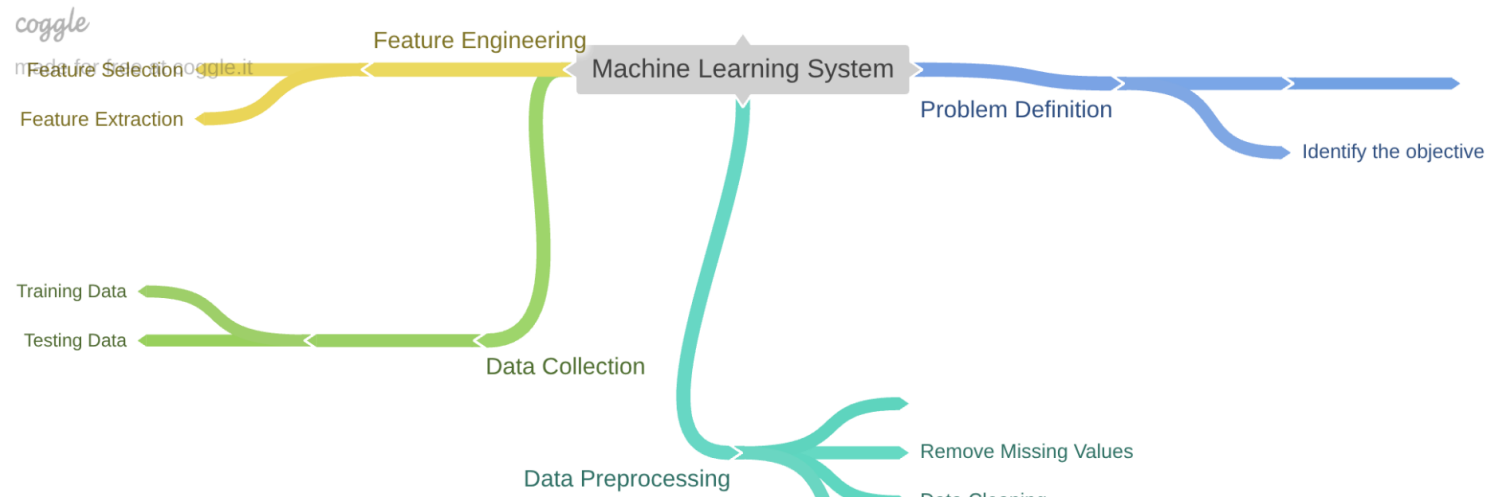


Introduction :

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed. Instead of writing rules manually, ML algorithms identify relationships in data and improve their performance through experience. A machine learning system consists of several stages,

starting from identifying the problem to making accurate predictions. Each stage plays an important role in improving the quality and reliability of the final model.

Concept Map :



1. Problem Definition

The first step is to clearly define the problem that needs to be solved. A well-defined problem helps in selecting the appropriate data and algorithm.

Example:

Predict whether an email is spam or not.

2. Data Collection

Relevant data is collected from different sources such as databases, sensors, websites, or surveys. The quality of the collected data directly affects the performance of the machine learning model.

Example:

Collect thousands of emails labeled as "Spam" and "Not Spam."

3. Data Preprocessing

Raw data often contains missing values, duplicate records, and noise. Data preprocessing cleans and transforms the data into a suitable format for training.

Common preprocessing tasks include:

- Removing missing values
- Eliminating duplicate records
- Normalizing data
- Encoding categorical values

Example:

Replace missing age values with the average age.

4. Feature Engineering

Feature engineering involves selecting or creating the most important features that influence the prediction. Better features usually lead to better model performance.

Example:

For house price prediction, important features include:

- Location
- Number of bedrooms
- Size of the house
- Age of the building

5. Algorithm Selection

After preparing the data, an appropriate machine learning algorithm is selected based on the problem type.

Supervised Learning

Supervised learning uses **labeled data**, where the correct output is already known. The model learns from input-output pairs.

Examples of supervised algorithms:

- Decision Tree
- Linear Regression
- Logistic Regression
- Support Vector Machine (SVM)

Example Application:

Predicting student exam results using previous academic records.

Unsupervised Learning

Unsupervised learning uses **unlabeled data**. The algorithm discovers hidden patterns or groups in the data without predefined outputs.

Examples of unsupervised algorithms:

- K-Means Clustering
- Hierarchical Clustering

Example Application:

Grouping customers based on their shopping behavior.

6. Model Training

During this stage, the selected algorithm learns from the training dataset by identifying relationships between input features and outputs.

Example:

A Decision Tree learns patterns from thousands of customer records to predict future purchases.

7. Model Validation

The trained model is tested using validation data to check whether it performs well on unseen data. Validation helps detect overfitting and underfitting.

Example:

Split the dataset into:

- 80% Training Data
- 20% Validation Data

8. Prediction / Decision

After successful evaluation, the trained model is deployed to make predictions on new data.

Example:

Predict whether a newly received email is Spam or Not Spam.

Conclusion

A machine learning system follows a structured pipeline consisting of problem definition, data collection, preprocessing, feature engineering, algorithm selection, model training, validation, evaluation, and prediction. Supervised learning is suitable when labeled data is available, while unsupervised learning is used to discover hidden patterns in unlabeled data. By following this pipeline, machine learning models can solve real-world problems efficiently and accurately.

Question 2

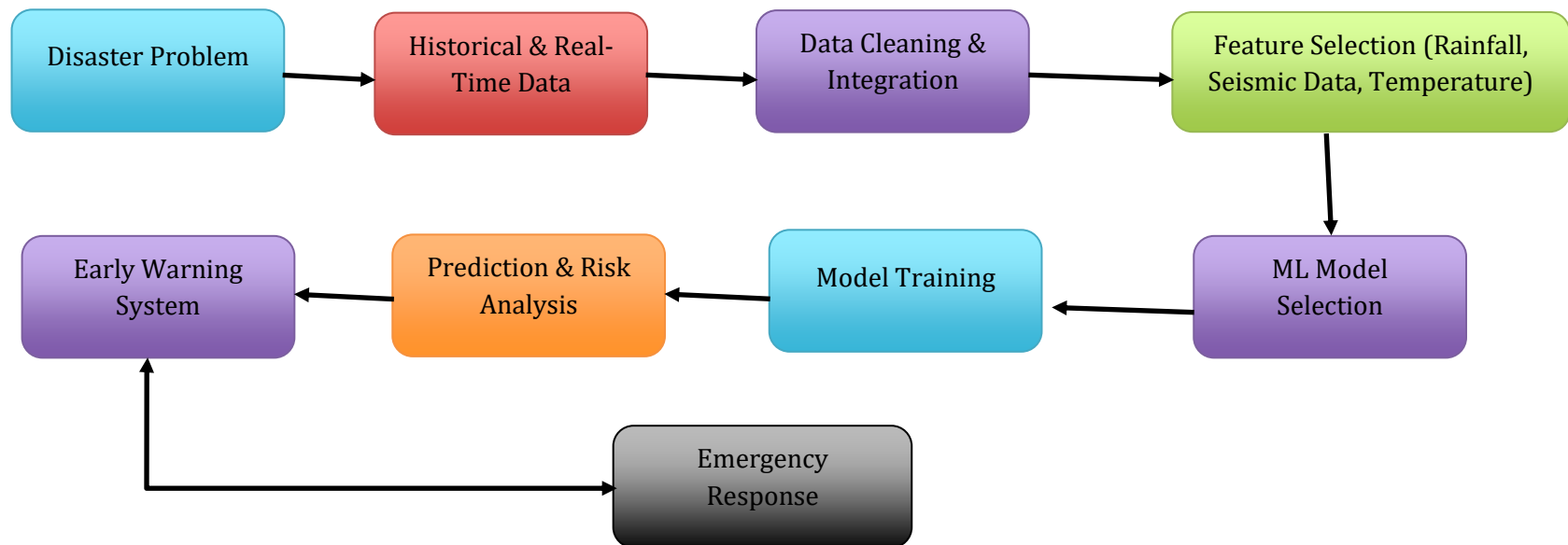
Central Concept: Machine Learning

Design a Machine Learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.

Flow chart :

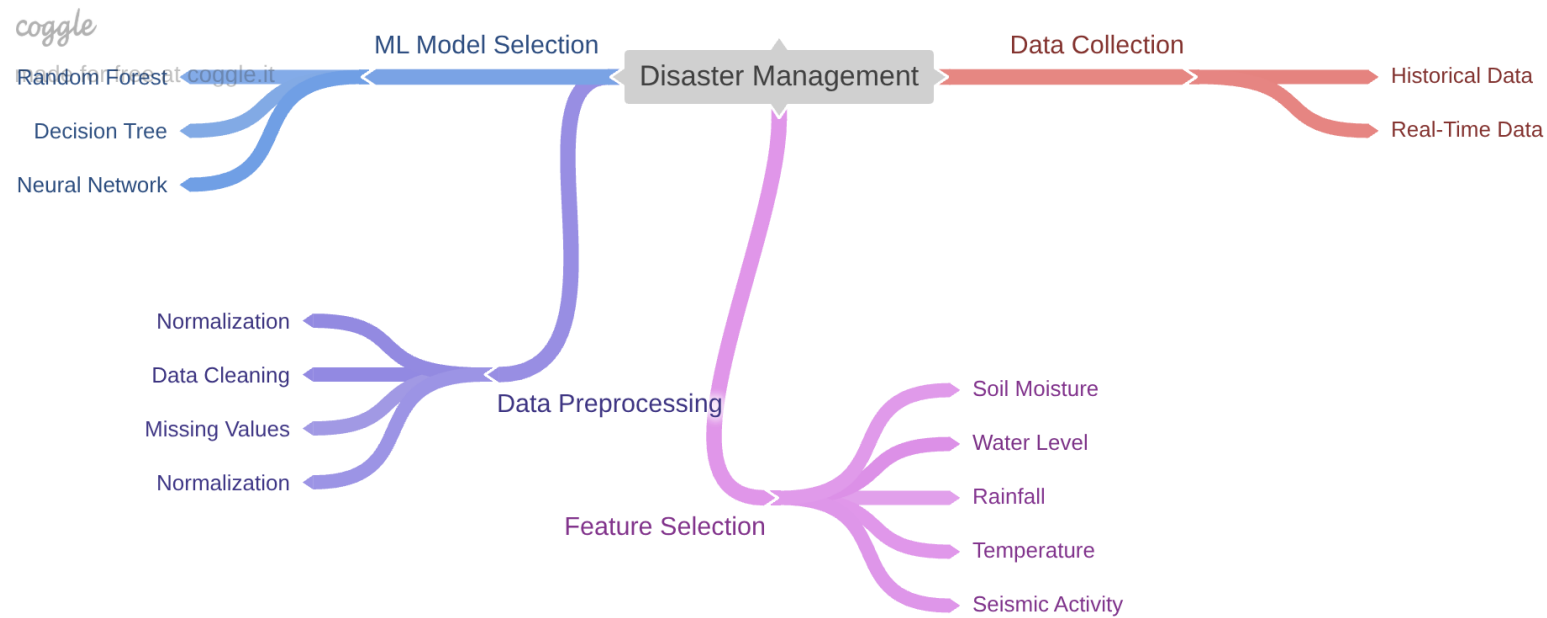


Introduction :

Disaster management is an important application of Machine Learning that helps predict natural disasters such as floods and earthquakes before they occur. By analyzing historical records and real-time sensor data, machine learning models can identify

patterns that indicate the possibility of a disaster. Accurate predictions help governments and emergency agencies take preventive measures, reduce damage, and save lives.

Concept Map :



Machine Learning System for Disaster Management

A disaster prediction system follows several stages to provide accurate and timely warnings.

1. Problem Definition

The objective is to predict disasters such as floods or earthquakes before they occur.

Example:

Predict whether heavy rainfall will cause flooding in a particular region.

2. Data Collection

The system collects both historical and real-time data.

Historical Data

- Previous flood records
- Earthquake history
- Weather reports
- River water levels

Real-Time Data

- Rainfall sensors
- Temperature sensors
- Water-level sensors
- Seismic sensors

- Satellite images

Combining historical and current information improves prediction accuracy.

3. Data Preprocessing

Raw data often contains errors and inconsistencies. Data preprocessing improves data quality before model training.

Data Cleaning

- Remove duplicate records
- Correct errors
- Handle missing values

Data Integration

Combine data collected from multiple sources into a single dataset.

Data Normalization

Scale different measurements into a common range so that no feature dominates the model.

4. Feature Selection

Feature selection identifies the most important factors affecting disaster prediction.

Important features include:

- Rainfall intensity

- River water level
- Temperature
- Humidity
- Soil moisture
- Wind speed
- Seismic activity

Selecting relevant features improves model accuracy and reduces computation time.

Choice of Machine Learning Model

Different machine learning algorithms can be used depending on the type of disaster and available data.

Decision Tree

- Easy to understand and interpret.
- Suitable for disaster classification.
- Fast prediction.

Example:

Predict whether flood risk is High, Medium, or Low.

(a) Steps Involved in Training, Validation, and Evaluation

Step 1 – Model Training

The selected algorithm learns from historical disaster data by identifying patterns between input features and disaster events.

Example:

Train the model using rainfall and river-level data collected over the past 20 years.

Step 2 – Model Validation

Validation checks the performance of the trained model using unseen data.

A common split is:

- 80% Training Data
- 20% Validation Data

Validation helps detect:

- Overfitting
- Underfitting

Step 3 – Model Evaluation

After validation, evaluate the model using performance metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score

A model with high accuracy and recall is preferred because it correctly identifies potential disasters while minimizing missed warnings.

(b) Early Warning System and Emergency Response

After evaluation, the model is deployed to monitor incoming real-time data continuously.

Early Warning System

The machine learning model predicts the likelihood of a disaster and sends alerts to authorities and the public.

Example:

If river water level and rainfall exceed safe limits, the system issues a flood warning.

Benefits

- Provides early alerts
- Reduces loss of life
- Minimizes property damage
- Helps people evacuate safely

Emergency Response

Once a disaster is predicted, emergency agencies can take immediate action.

Actions include:

- Evacuating people from high-risk areas
- Deploying rescue teams
- Distributing food and medical supplies
- Managing traffic and shelters

- Coordinating disaster relief operations

This improves response time and reduces the overall impact of the disaster.

Conclusion

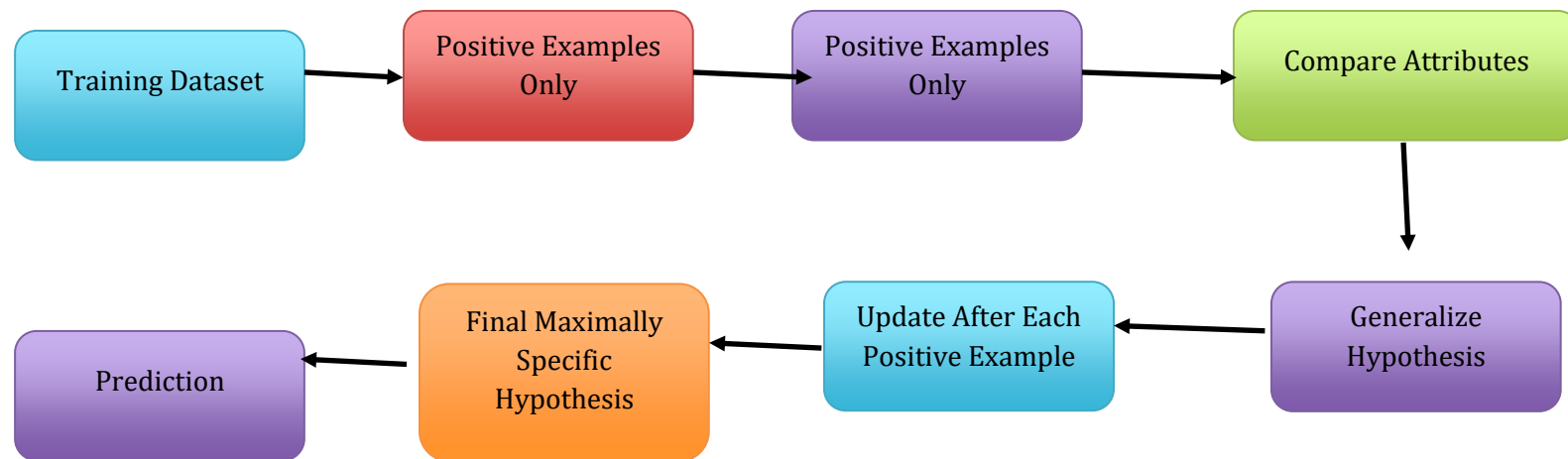
Machine Learning plays a significant role in disaster management by analyzing historical records and real-time sensor data to predict floods, earthquakes, and other natural disasters. Data preprocessing and feature selection improve the quality of predictions, while algorithms such as Decision Trees, Random Forests, and Neural Networks provide reliable forecasting. A well-designed machine learning system supports early warning systems and emergency response, helping save lives and reduce damage.

Question 3

Central Concept: Machine Learning

In the context of the Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis.

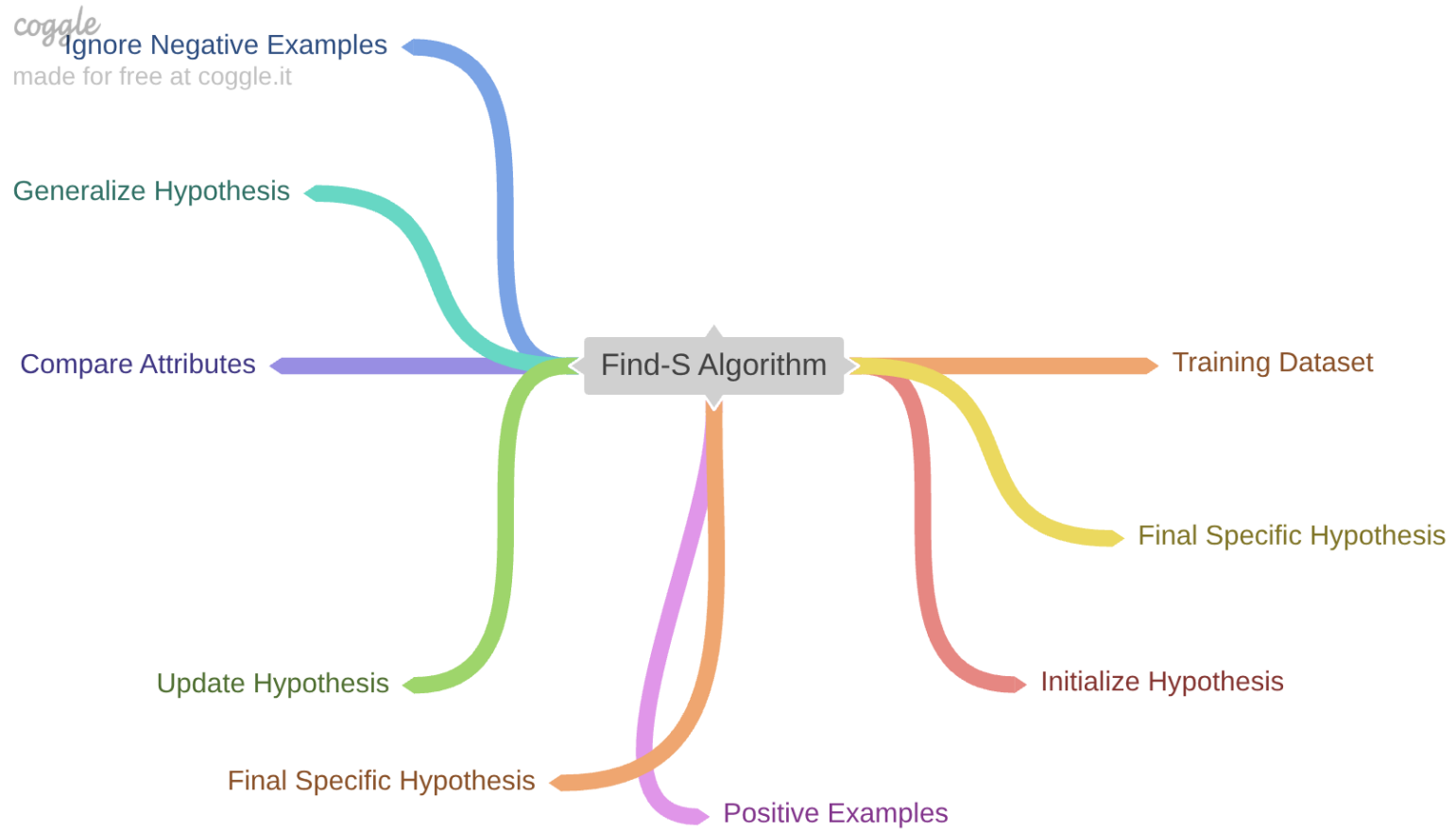
Flow Chart :



Introduction :

The Find-S Algorithm is one of the simplest concept learning algorithms in Machine Learning. It is used to determine the most specific hypothesis that is consistent with all the positive training examples. The algorithm completely ignores negative examples and updates the hypothesis whenever differences are found among positive examples. Find-S is mainly used in supervised learning for concept learning problems.

Concept Map :



Find-S Algorithm

Steps of the Find-S Algorithm

1. Initialize the hypothesis with the **most specific hypothesis ($\emptyset$)**.
2. Consider only **positive training examples**.
3. Compare the current hypothesis with each positive example.
4. If an attribute differs, replace it with ? (any value).
5. Ignore all negative examples.
6. Repeat until all positive examples are processed.
7. The resulting hypothesis is the **maximally specific hypothesis**.

Given Dataset

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes

4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Step-by-Step Solution

Initial Hypothesis

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Example 3 (Positive)

$S_1 = \langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$

Example 4 (Positive)

Compare with previous hypothesis.

$S_2 = \langle ?, ?, \text{High}, \text{Weak} \rangle$

Example 5 (Positive)

Humidity changes from High to Normal.

$S_3 = \langle ?, ?, ?, \text{Weak} \rangle$

Example 7 (Positive)

Wind changes from Weak to Strong.

$S_4 = \langle ?, ?, ?, ? \rangle$

(All negative examples are ignored.)

Final Hypothesis

$H = \langle ?, ?, ?, ? \rangle$

The final hypothesis shows that all attributes became generalized because the positive examples had different values for every attribute.

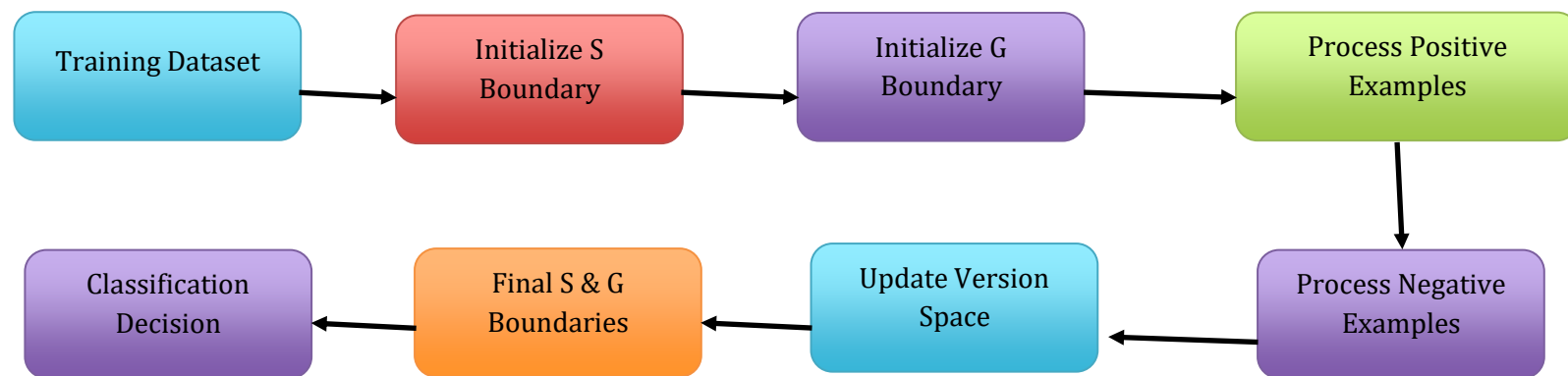
Conclusion

The Find-S algorithm learns from **positive examples only** and ignores negative examples. In this dataset, the hypothesis gradually became more general as different positive examples were processed. The final hypothesis is $\langle ?, ?, ?, ? \rangle$, indicating that no single attribute remained specific across all positive examples.

Question 4

Central Concept: Machine Learning

In the context of the Candidate Elimination algorithm, determine the version space by computing the Specific (S) and General (G) boundaries. Show the step-by-step updates after processing each training example.



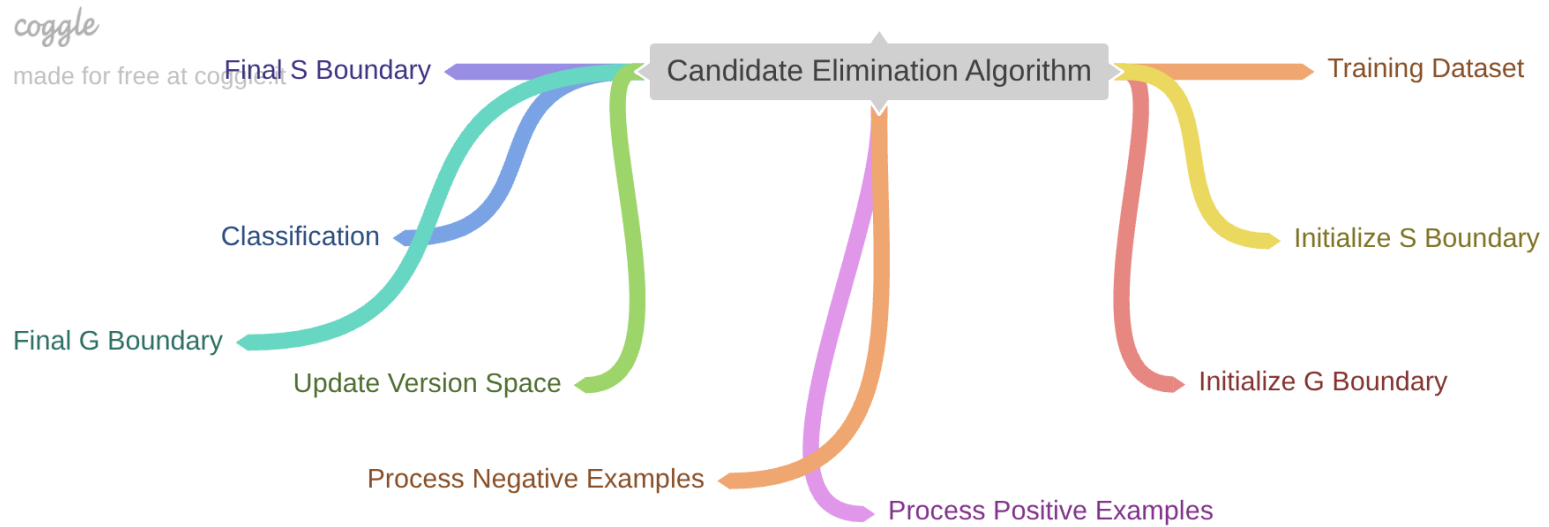
Introduction :

The Candidate Elimination Algorithm is a supervised learning algorithm used to find all hypotheses that are consistent with the training data. It maintains two boundaries:

- Specific Boundary (S): Most specific hypothesis.
- General Boundary (G): Most general hypothesis.

As each training example is processed, the algorithm updates both boundaries to form the Version Space.

Concept Map :



Given Dataset

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes

3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

Step-by-Step Solution

Step 1: Initialize

Specific Boundary (S)

$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

General Boundary (G)

$\langle ?, ?, ?, ? \rangle$

Step 2: Process Example 1 (Positive)

Update S with the first positive example.

S = <Sunny, Warm, Normal, Weak>

G = <?, ?, ?, ?>

Step 3: Process Example 2 (Positive)

Compare with S.

Humidity and Wind are different, so generalize them.

S = <Sunny, Warm, ?, ?>

G = <?, ?, ?, ?>

Step 4: Process Example 3 (Negative)

Specialize G to exclude the negative example while remaining consistent with S.

S = <Sunny, Warm, ?, ?>

G = <Sunny, ?, ?, ?>

Step 5: Process Example 4 (Positive)

Temperature changes from Warm to Cold.

S = <Sunny, ?, ?, ?>

G = <Sunny, ?, ?, ?>

Step 6: Process Example 5 (Negative)

The negative example does not satisfy the current S boundary, so no further change is required.

Final Version Space

Specific Boundary (S)

<Sunny, ?, ?, ?>

General Boundary (G)

<Sunny, ?, ?, ?>

Conclusion

The Candidate Elimination algorithm updates both the **Specific (S)** and **General (G)** boundaries after each training example. It considers both positive and negative examples to determine the **Version Space**. For this dataset, the final boundaries are:

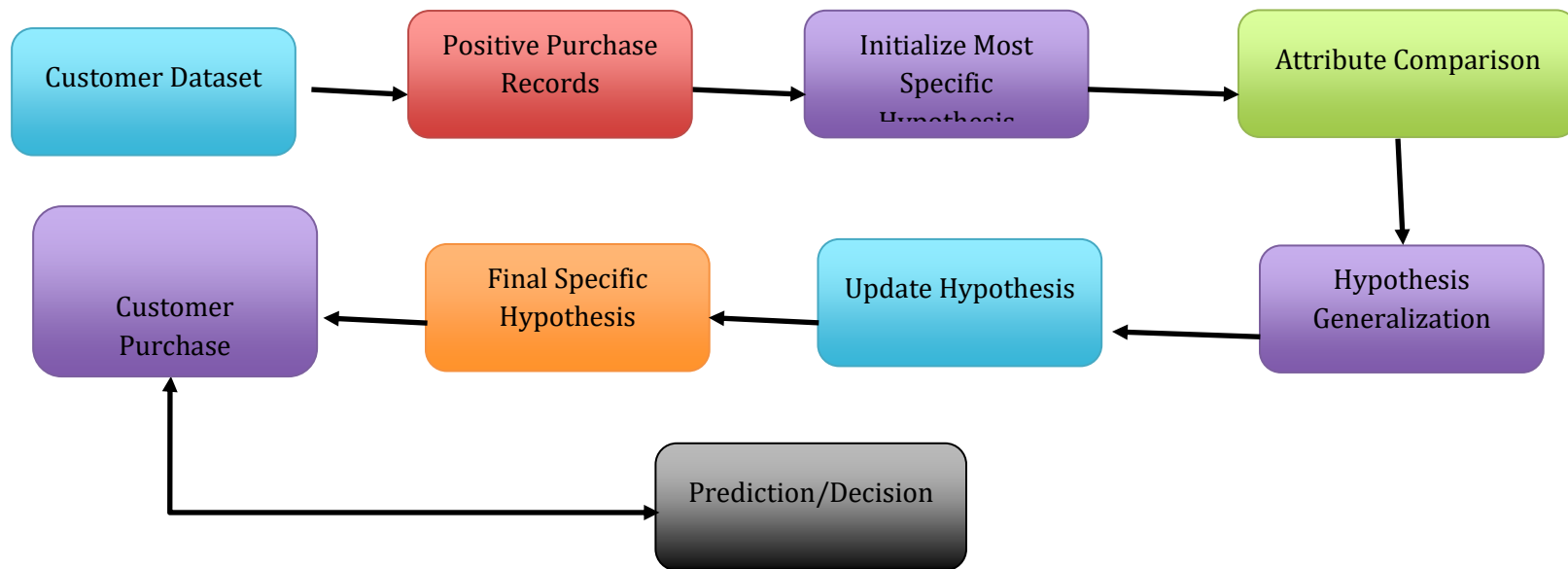
- **S** = <Sunny, ?, ?, ?>
- **G** = <Sunny, ?, ?, ?>

These hypotheses correctly classify the given training examples.

Question 5

Central Concept: Machine Learning

In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.



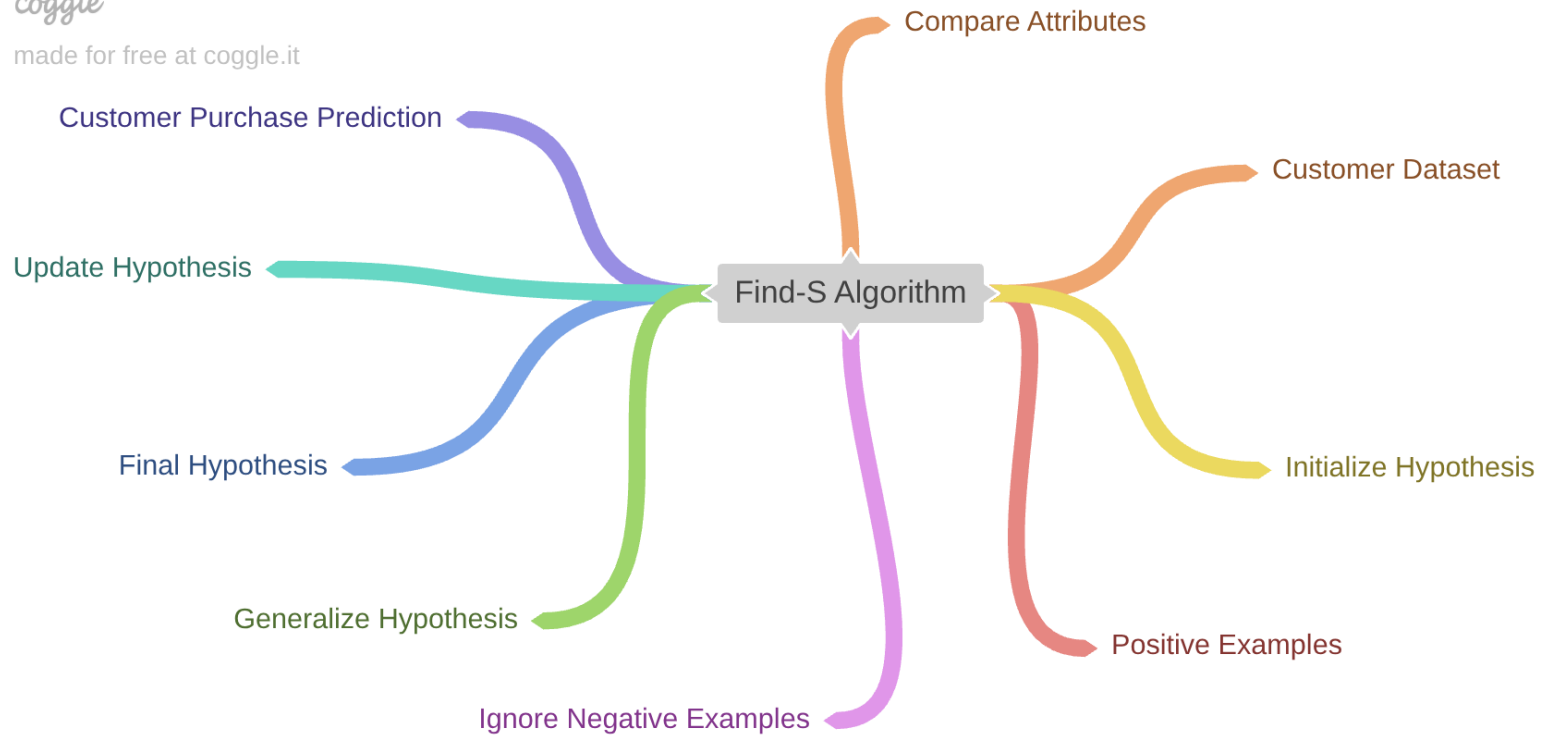
Introduction :

The Find-S Algorithm is a supervised learning algorithm that identifies the most specific hypothesis by considering only positive training examples. It starts with the most specific hypothesis and generalizes it whenever differences are found among positive examples. Negative examples are ignored.

Concept map :

coggle

made for free at coggle.it



Given Dataset

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

Step-by-Step Solution

Step 1: Initialize Hypothesis

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Step 2: Example 1 (Negative)

Ignored

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Step 3: Example 2 (Negative)

Ignored

$$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

Step 4: Example 3 (Positive)

First positive example becomes the hypothesis.

$$S_1 = \langle \text{Middle, High, No, Fair} \rangle$$

Step 5: Example 4 (Positive)

Compare with S_1 :

- **Age:** Middle $\rightarrow$ Senior $\rightarrow$?
- **Income:** High $\rightarrow$ Medium $\rightarrow$?
- **Student:** No $\rightarrow$ No $\rightarrow$ **No**
- **Credit Rating:** Fair $\rightarrow$ Fair $\rightarrow$ **Fair**

Updated Hypothesis:

$$S_2 = \langle ?, ?, \text{No, Fair} \rangle$$

Final Hypothesis

$$H = \langle ?, ?, \text{No, Fair} \rangle$$

This hypothesis indicates that a customer is likely to buy the product if the **Student = No** and **Credit Rating = Fair**, while **Age** and **Income** can be of any value.

Conclusion

The Find-S algorithm processes only positive examples to find the most specific hypothesis. In this dataset, the final hypothesis is **<?, ?, No, Fair>**, which satisfies all the positive training examples and can be used to predict future customer purchases.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand